



CMR COLLEGE OF ENGINEERING & TECHNOLOGY
(UGC AUTONOMOUS)

B.Tech III Semester Supplementary Examinations July-2025

Course Name: Analog & Digital Electronics

(Common for CSE & IT)

Date: 03.07.2025 FN

Time: 3 hours

Max.Marks: 70

(Note: Assume suitable data if necessary)

PART-A

Answer all TEN questions

Each question carries TWO marks.

10x2=20M

- | | |
|--|-----|
| 1. Draw the characteristics of PN-Diode. | 2 M |
| 2. Define ripple factor of rectifier. | 2 M |
| 3. How the transistor acts as an amplifier? | 2 M |
| 4. Draw the characteristics of UJT. | 2 M |
| 5. What are the different types of modes in MOSFET? | 2 M |
| 6. Define Number system. | 2 M |
| 7. What is meant by combinational circuits? | 2 M |
| 8. Why NAND and NOR gates are called as a universal Gates? | 2 M |
| 9. Define counter. | 2 M |
| 10. What is meant by sequential circuits? | 2 M |

PART-B

Answer the following. Each question carries TEN Marks.

5x10=50M

- | | |
|--|-----|
| 11.A). Explain the operation of Zener Diode. | 10M |
| OR | |
| 11. B). Explain the break down mechanism of a semiconductor diode. | 10M |
| 12. A). With a neat sketches explain the current components of a transistor. | 10M |
| OR | |
| 12. B). Draw the input and output characteristics of a Transistor in CE-Configuration. | 10M |
| 13. A). Explain the construction and working of JFET. | 10M |
| OR | |
| 13. B). With a suitable example convert the 4-bit binary to gray code. | 10M |
| 14. A). Explain the working principle of Encoder and Decoder. | 10M |
| OR | |
| 14. B). Explain the working principle of Multiplexer? | 10M |
| 15. A). Design a 3-bit binary up-counter using a JK-Flipflop | 10M |
| OR | |
| 15. B). i) Conversion of SR-Flip flop to JK-Flip flop. | 5M |
| ii) Conversion of JK-Flip flop to D-Flip flop. | 5M |
